Tools, Visualization, and Experimental Evaluation

CSE4107: Artificial Intelligence

Logistics

- **Instructor:** Raihan Tanvir
  - ✓ Lecturer, Department of CSE, AUST
  - ✓ Email: raihantanvir.cse@aust.edu
  - ✓ Office Hours: TBA
  - ✓ Website: https://raihan-tanvir.github.io/
- **Semester**: Spring 2022

References

- Some of the examples, notes, definitions, ideas are taken from
  - courses.cs.washington.edu › cse446 › slides › intro (University of Washington)
  - http://www2.cs.uh.edu/~ceick/ [CHRISTOPH F. EICK (University of Houston)]
  - Medium
  - AT&T Laboratories, Cambridge UK [http://www.uk.research.att.com/facedatabase.html]
  - Several other internet repositories
  - https://acutecaretesting.org/en/articles/roc-curves-what-are-they-and-how-are-they-used
  - https://towardsdatascience.com/understanding-auc-roc-curve-68b2303cc9c5
  - https://en.wikipedia.org/wiki/Receiver_operating_characteristic

Performance Metrics (PM)

Confusion Matrix

- a performance measurement for machine learning classification problem
- output can be two or more classes
- a table with **four** different combinations of **predicted** and **actual** values

Predicted Values \ Actual Values	+ve (1)	-ve (0)
+ve (1)	**TP**	**FP**
-ve (0)	**FN**	**TN**

PM — Confusion Matrix

- **True Positives (TP)** − It is the case when both actual class & predicted class of data point is 1.
- **True Negatives (TN)** − It is the case when both actual class & predicted class of data point is 0.
- **False Positives (FP)** − It is the case when actual class of data point is 0 & predicted class of data point is 1.
- **False Negatives (FN)** − It is the case when actual class of data point is 1 & predicted class of data point is 0.

Actual Values

Predicted Values	+ve (1)	-ve (0)
+ve (1)	**TP**	**FP**
-ve (0)	**FN**	**TN**

PM — Confusion Matrix

Example

- **1**. a person is having a cancer
- **0**. a person is not having a cancer

✓ **True Positives (TP)** − a person is actually having cancer (1) and the model classifying his case as cancer (1) comes under *True positive.*

✓ **True Negatives (TN)** − a person NOT having cancer and the model classifying his case as Not cancer comes under *True Negatives.*

✓ **False Positives (FP)** − a person NOT having cancer and the model classifying his case as cancer comes under *False Positives.*

✓ **False Negatives (FN)** − a person having cancer and the model classifying his case as No-cancer comes under *False Negatives.*

PM — Accuracy

- most common performance metric for classification algorithms.
- defined as the number of correct predictions made as a ratio of all predictions made

- **Accuracy is a good measure when the target variable classes in the data are nearly balanced**
- Accuracy should **NEVER** be used as a measure when the target variable classes in the data are a majority of one class.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

PM — Precision/PPV

- defined as the number of correct documents returned by our ML model
- Ex. - Precision is a measure that tells us what proportion of patients that we diagnosed as having cancer, actually had cancer.

$$Precision = \frac{TP}{TP + FP}$$

In our cancer example with 100 people, only 5 people have cancer. Let's say our model is very bad and predicts every case as **Cancer**. Since we are predicting everyone as having cancer, our denominator (True positives and False Positives) is 100 and the numerator, person having cancer and the model predicting his case as cancer is 5. So in this example, we can say that **Precision** of such model is 5%. **[medium]**

PM — Recall/Sensitivity/TPR

- defined as the number of positives returned by our ML model
- Ex. - a measure that tells us what proportion of patients that actually had cancer was diagnosed by the algorithm as having cancer.

$$Recall = \frac{TP}{TP + FN}$$

In our cancer example with 100 people, 5 people actually have cancer. Let's say that the model predicts every case as cancer. So our denominator (True positives and False Negatives) is 5 and the numerator, person having cancer and the model predicting his case as cancer is also 5 (Since we predicted 5 cancer cases correctly). So in this example, we can say that the **Recall** of such model is 100%. And Precision of such a model (As we saw above) is 5%

PM — Precision & Recall

- recall gives us information about a classifier's performance with respect to false negatives (how many did we miss), while precision gives us information about its performance with respect to false positives (how many did we caught).
- **Precision:** It is about being precise. So even if we managed to capture only one cancer case, and we captured it correctly, then we are 100% precise.
- **Recall:** It is not so much about capturing cases correctly but more about capturing all cases that have "cancer" with the answer as "cancer". So if we simply always say every case as "cancer", we have 100% recall.

PM — Specificity/Selectivity/TNR

- contrast to recall
- defined as the number of negatives returned by our ML model.
- it is a measure that tells us what proportion of patients that did NOT have cancer, were predicted by the model as non-cancerous.

$$Specificity = \frac{TN}{TN + FP}$$

PM — F1 Score

- Harmonic mean of precision and recall
- F1 score is the weighted average of the precision and recall

- difficult to compare two models with low precision and high recall or vice versa.
- to make them comparable, we use F-Score.
- F-score helps to measure Recall and Precision at the same time.

$$F - measure = \frac{2 * precision * recall}{precision + recall}$$

PM — R-squared

- It is a statistical measure between 0 and 1
- Calculates how similar a regression line is to the data it's fitted to
- If it's a 1, the model 100% predicts the data variance; if it's a 0, the model predicts none of the variance.

$$R - squared = \frac{explained\ variance\ of\ the\ model}{total\ variance\ of\ the\ target\ variable}$$

PM — R-squared

$$R - squared = 1 - \frac{S_{res}^2}{S_{total}^2}$$

- S_{res}^2 : sum of squares of the residual errors
- S_{total}^2 : total sum of the errors

Let $R^2 = 0.70$
It can be referred that 70% of the changeability of the dependent output attribute can be explained by the model while the remaining 30 % of the variability is still unaccounted for.

PM — R-squared

$$R - squared = 1 - \frac{S_{res}^2}{S_{total}^2}$$

Case 1 Model gives accurate results

Actual (y)	Predicted(f)	Error (E1=y-f)	E1^2	E2=y-Mean	E2^2
10	10	0	0	-10	100
20	20	0	0	0	0
30	30	0	0	10	100
Mean= 20			**Sum=0**		**Sum=200**

$$R - squared = 1 - \frac{0}{200} = 1$$

Case 2 Model gives same results always

Actual (y)	Predicted(f)	Error (E1=f-y)	E1^2	E2=y-Mean	E2^2
10	20	10	100	-10	100
20	20	0	0	0	0
30	20	-10	100	10	100
Mean= 20			**Sum=200**		**Sum=200**

$$R - squared = 1 - \frac{200}{200} = 0$$

Case 3 Model gives ambiguous results

Actual (y)	Predicted(f)	Error (E1=y-f)	E1^2	E2=y-Mean	E2^2
10	30	-20	400	-10	100
20	10	10	100	0	0
30	20	10	100	10	100
Mean= 20			**Sum=600**		**Sum=200**

$$R - squared = 1 - \frac{600}{200} = -2$$

PM — Summary

$$R - squared = 1 - \frac{S_{res}^2}{S_{total}^2}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}|$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2}$$

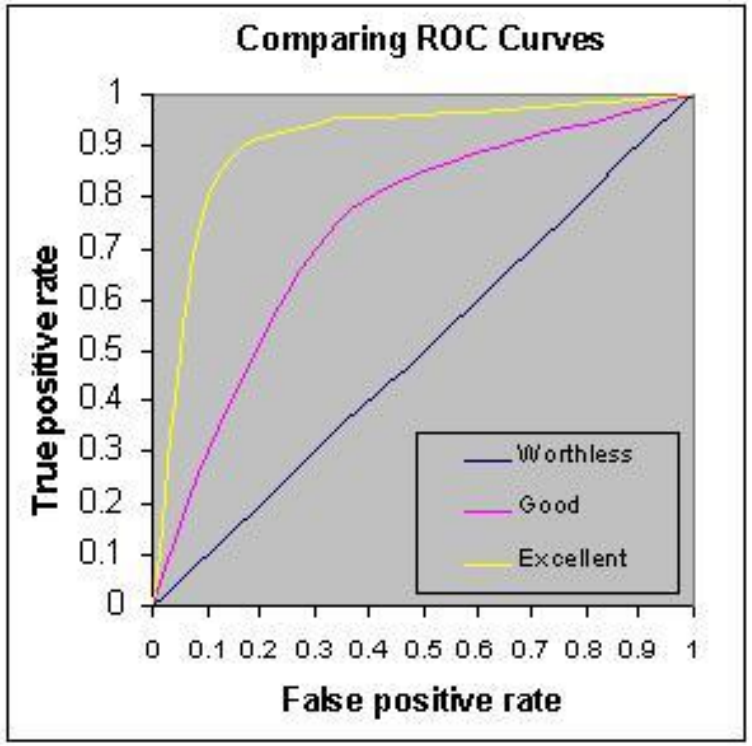
When?
Why?
Where?
Differences?

Exercise

A	B	C	C′
TP=63 FP=28 91 FN=37 TN=72 109 100 100 200	TP=77 FP=77 154 FN=23 TN=23 46 100 100 200	TP=24 FP=88 112 FN=76 TN=12 88 100 100 200	TP=76 FP=12 88 FN=24 TN=88 112 100 100 200
TPR = 0.63	TPR = 0.77	TPR = 0.24	TPR = 0.76
FPR = 0.28	FPR = 0.77	FPR = 0.88	FPR = 0.12
PPV = 0.69	PPV = 0.50	PPV = 0.21	PPV = 0.86
F1 = 0.66	F1 = 0.61	F1 = 0.23	F1 = 0.81
ACC = 0.68	ACC = 0.50	ACC = 0.18	ACC = 0.82

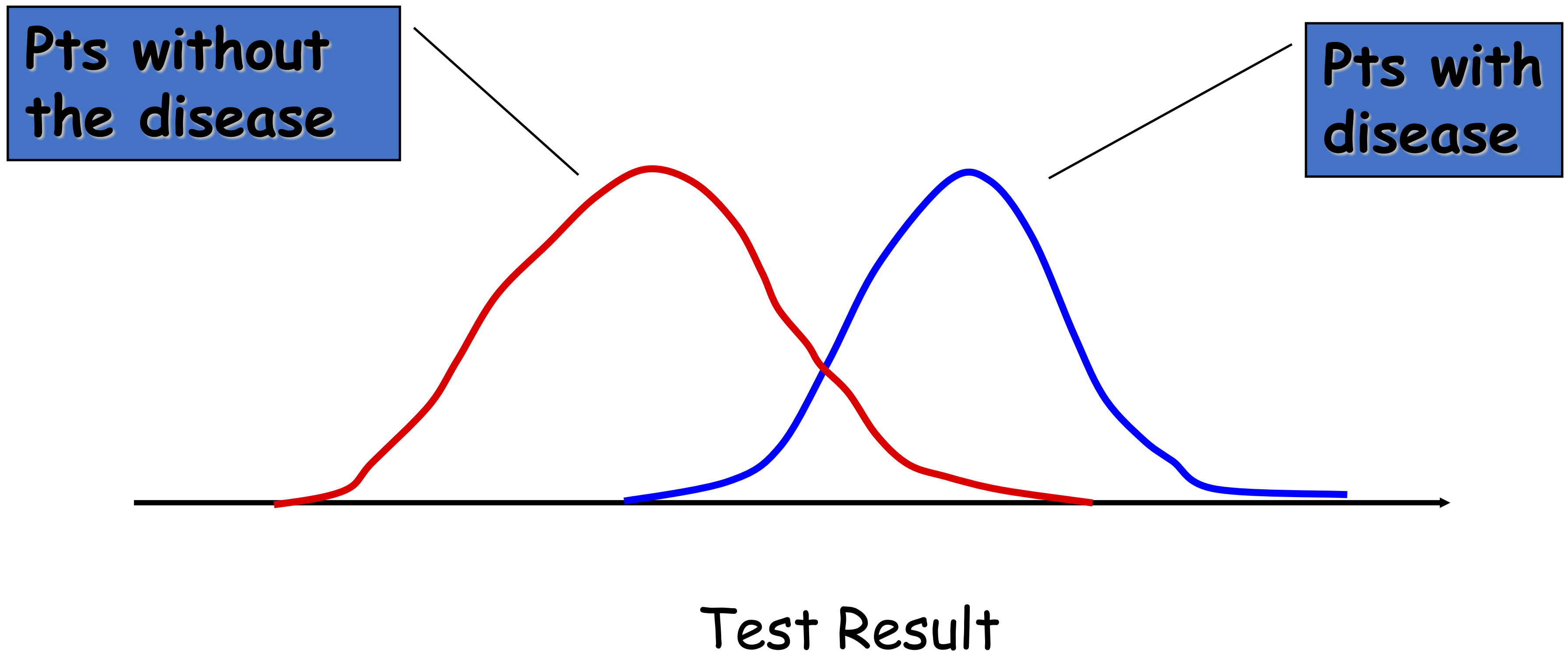
AUC-ROC Curve

- AUC - ROC curve is a performance measurement for classification problem at various thresholds settings
- ROC is a probability curve and AUC represents degree or measure of separability
- It tells how much model is capable of distinguishing between classes
- Higher the AUC, better the model is at predicting 0s as 0s and 1s as 1s
- So, Higher the AUC, better the model is at distinguishing between patients with disease and no disease.

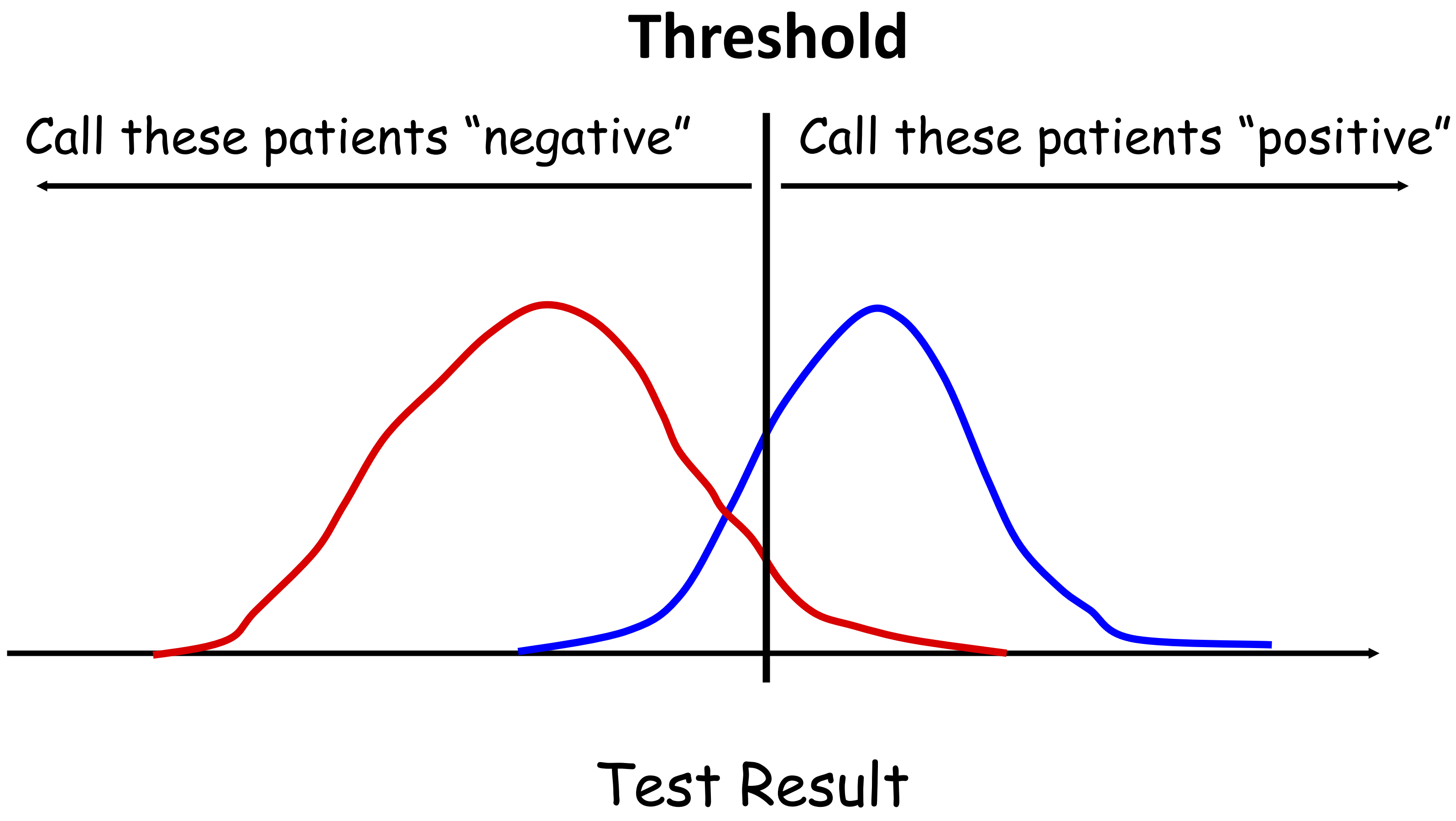


https://acutecaretesting.org/en/articles/roc-curves-what-are-they-and-how-are-they-used
https://towardsdatascience.com/understanding-auc-roc-curve-68b2303cc9c5
https://en.wikipedia.org/wiki/Receiver_operating_characteristic

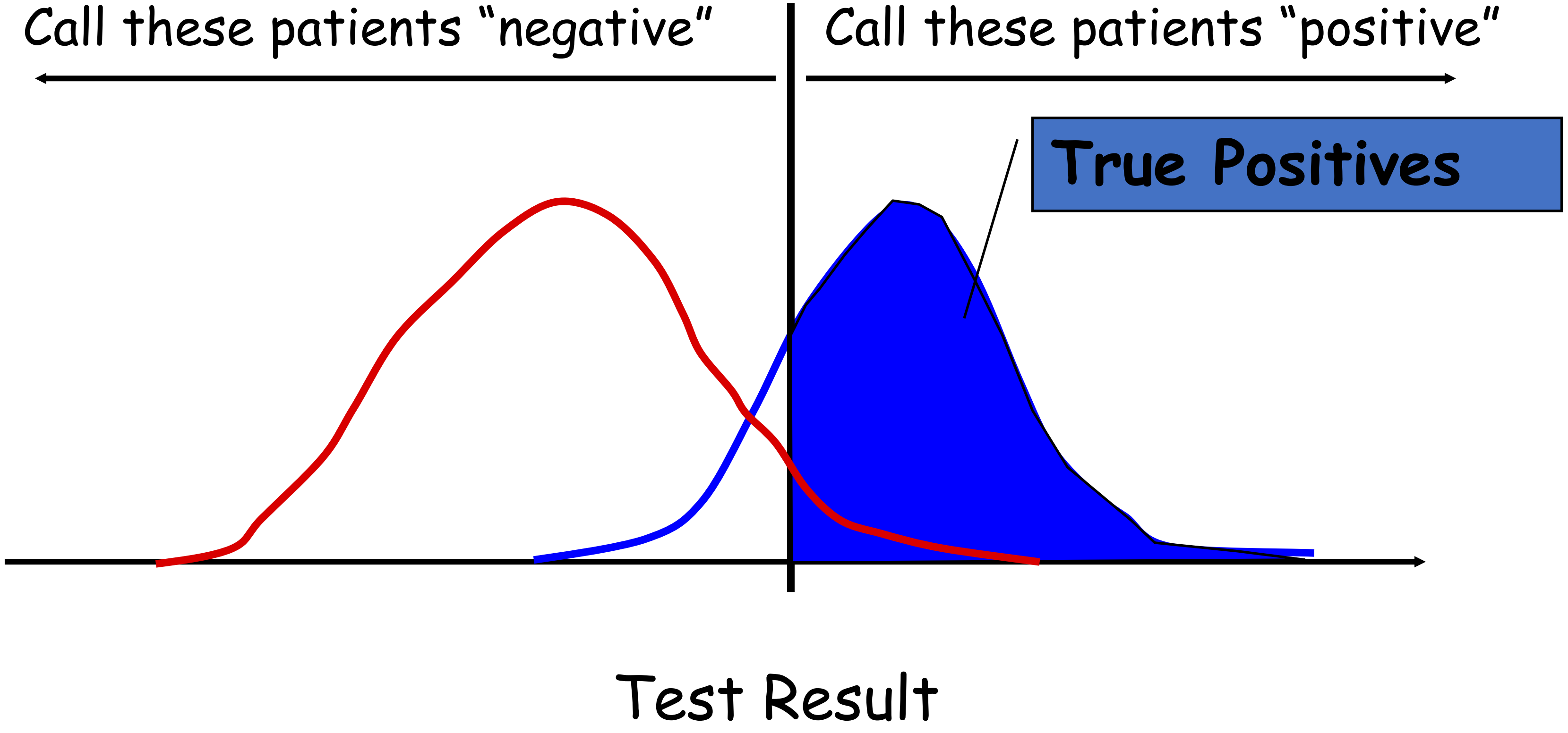
ROC Curve



ROC Curve

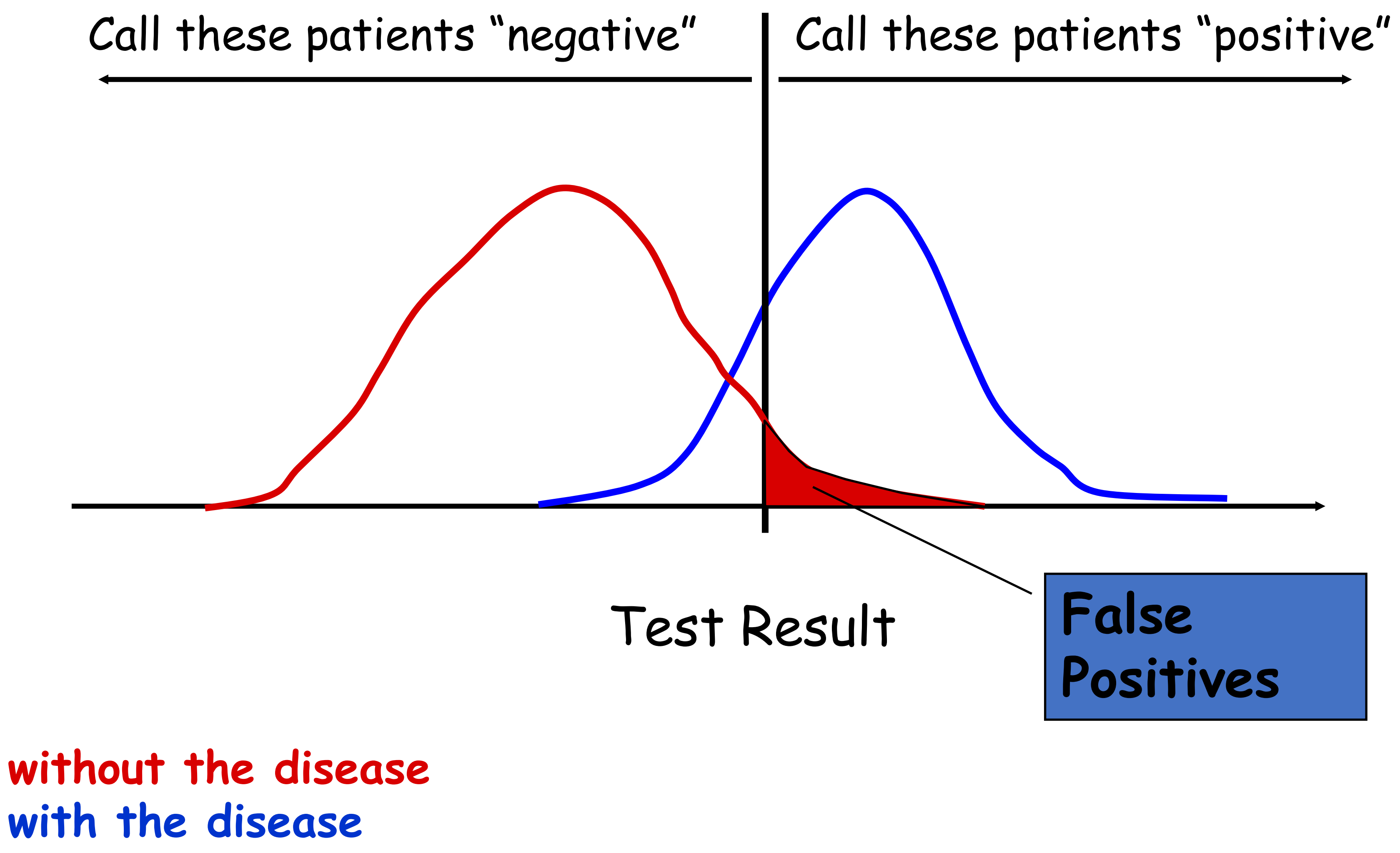

Threshold
Call these patients "negative"
Call these patients "positive"
Test Result

ROC Curve

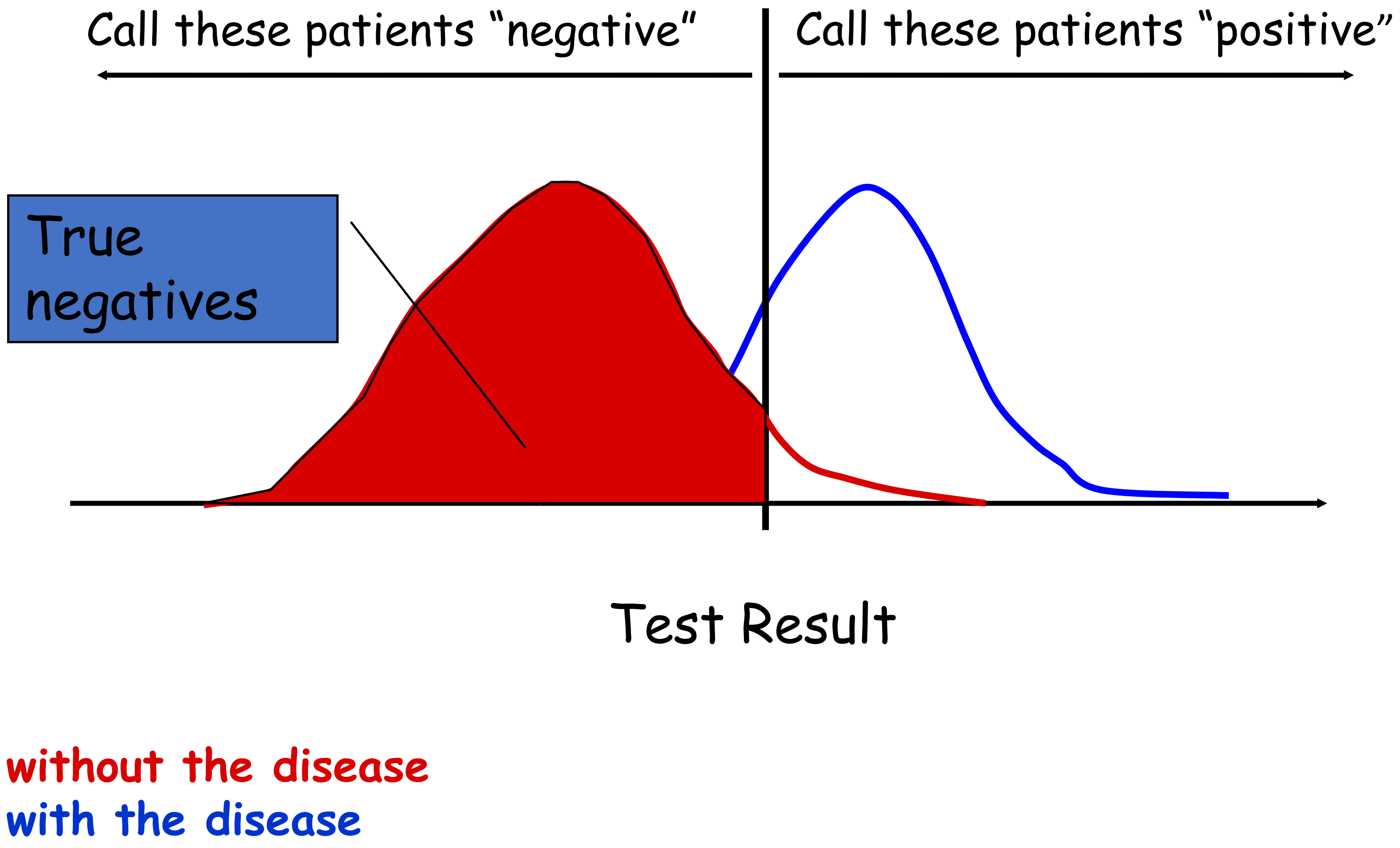


without the disease
with the disease

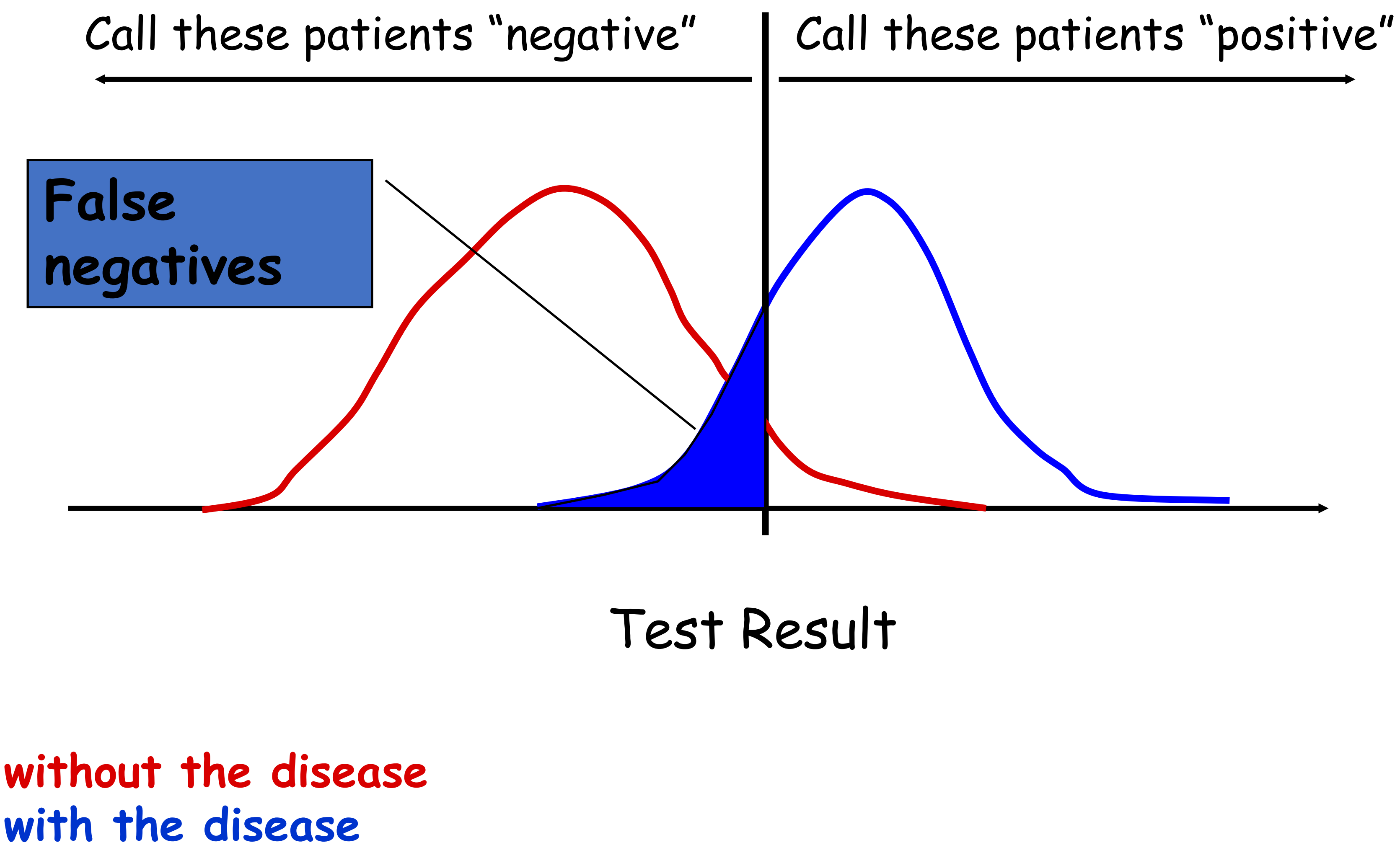
AUC-ROC Curve



AUC-ROC Curve



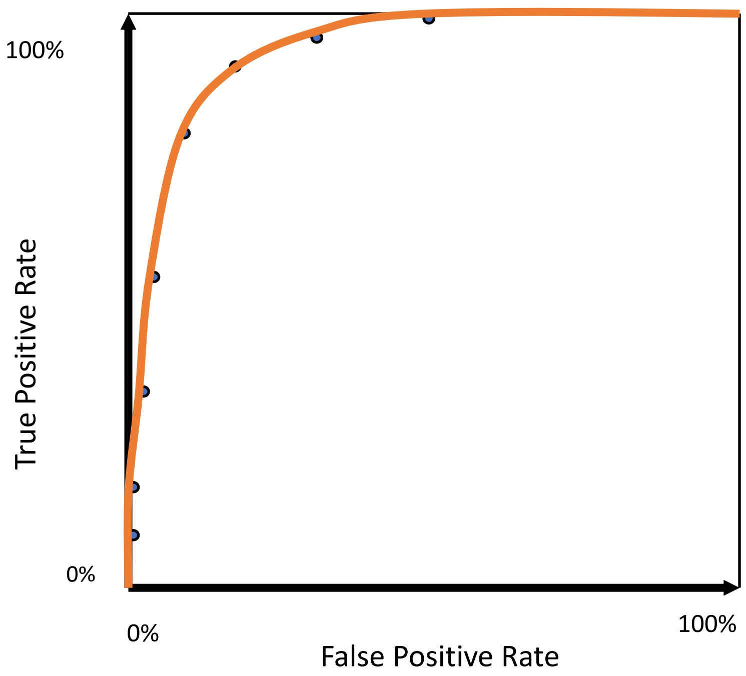
AUC-ROC Curve



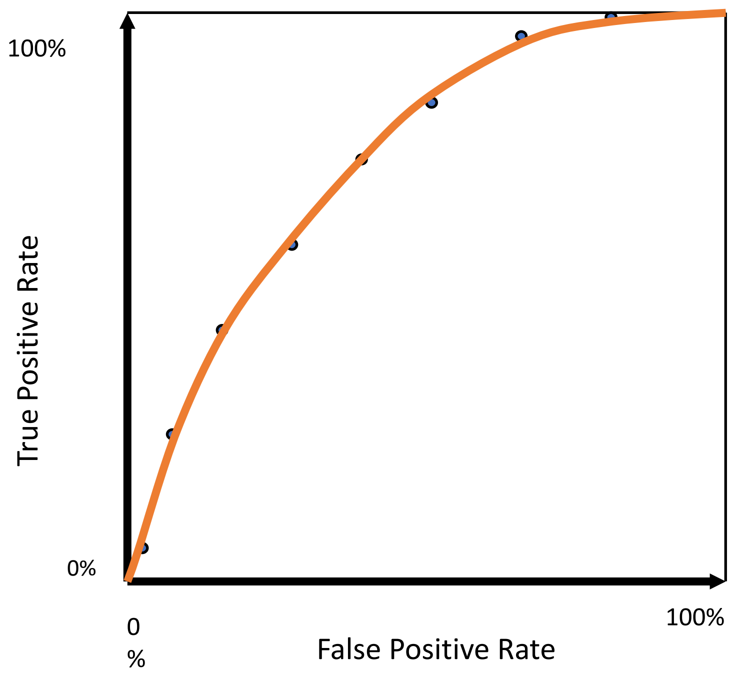
ML

ROC Curve Comparison

A good test:

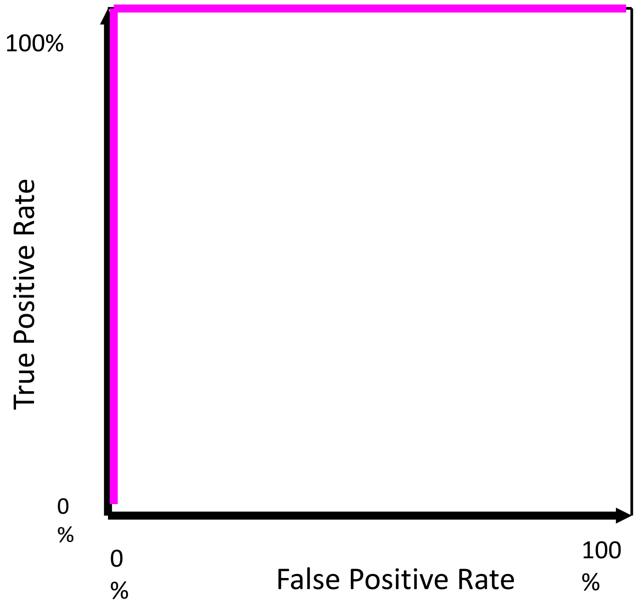


A poor test:



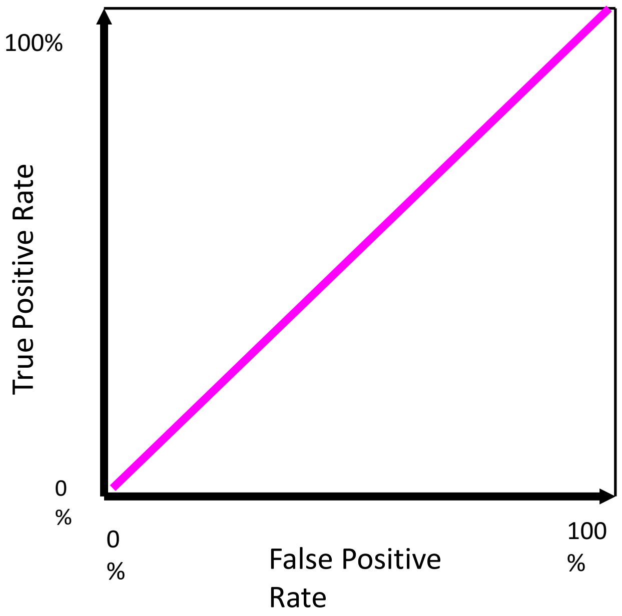
ROC Curve Comparison

Best Test:



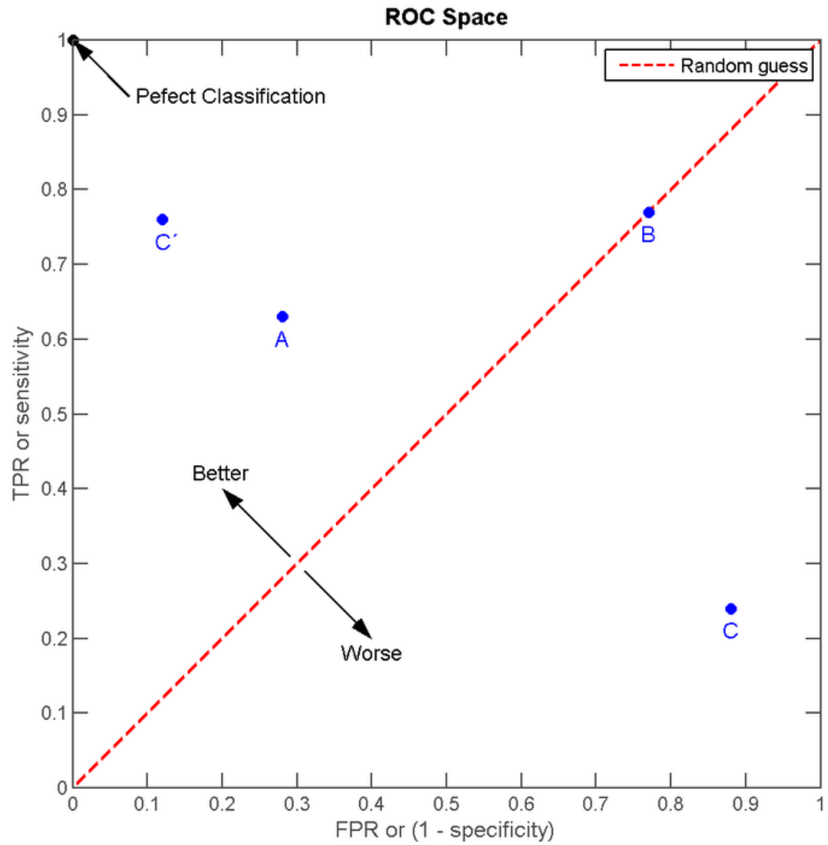
The distributions don't overlap at all

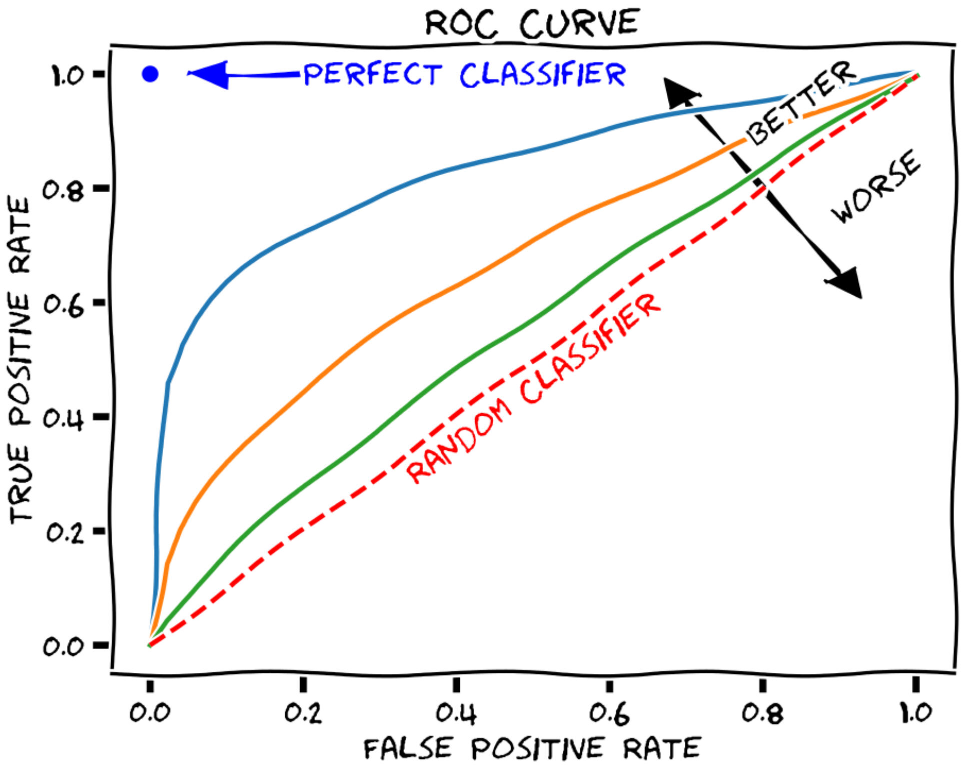
Worst test:



The distributions overlap completely

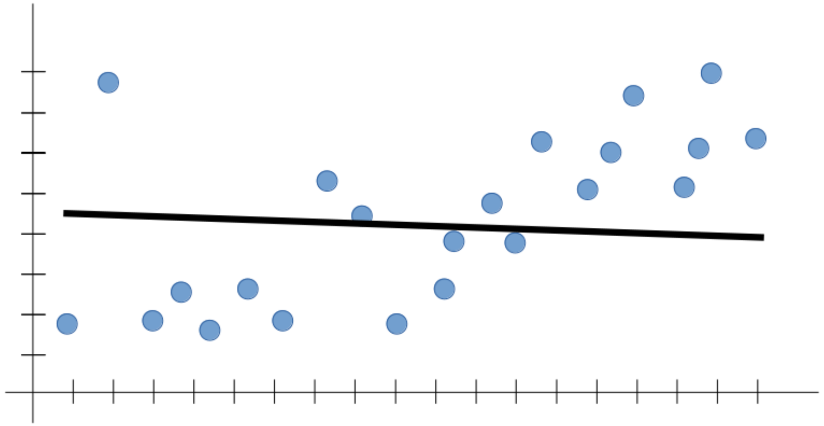
ROC Curve Summary

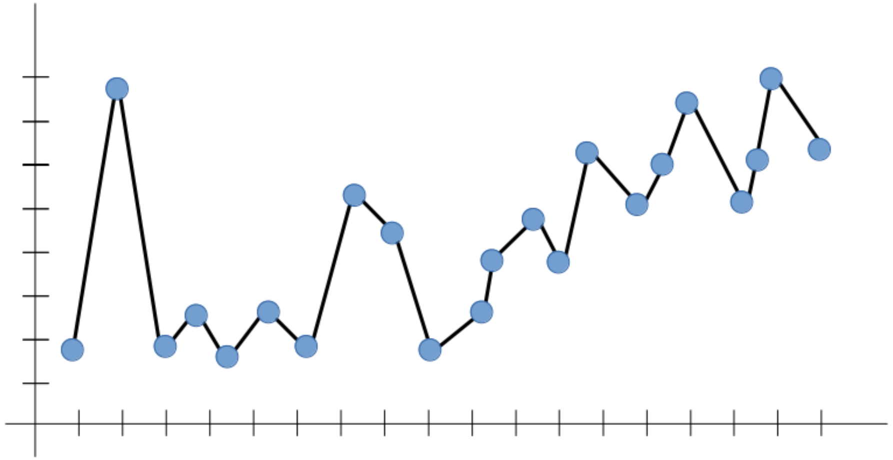




Training Stage

Underfitting





Overfitting